Distributed Quantum-Assisted Robust AoII Minimization in Satellite-Ground Integrated Edge Networks

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***Abstract*—Edge servers that act on a monitored process need an estimate that is correct, not merely recent. In satellite-ground integrated networks (SAGIN), satellite handover and shadowing interrupt links for long enough that the monitored process can change state several times, so the age of incorrect information (AoII) is the appropriate measure of estimation quality. This paper formulates the robust, network-level AoII minimization problem VANGUARD, in which update rates, serving-node association, and bandwidth are chosen jointly to minimize the worst-case time-average AoII over a set of disruption scenarios, and solves it with SENTINEL, a distributed hybrid quantum-classical procedure. A renewal-interval decomposition turns the history-dependent AoII into a closed-form cost per inter-delivery interval. The minimax over scenarios is dualized into scenario weights updated by a classical outer loop, so that each satellite-ground domain solves a linear assignment subproblem encoded as a compact quadratic unconstrained binary optimization (QUBO) whose one-hot structure is enforced by XY mixers rather than penalties; the domains are coordinated through the sharing form of ADMM, and every returned schedule is certified by its exact worst-case AoII. The evaluation uses Starlink ephemerides, the 3GPP TR 38.811 non-terrestrial channel, and acknowledgement feedback for all policies, with an exact integer-programming reference and paired tests over 30 to 60 seeds. The certificate held in 98.7% of orchestration epochs under real passes; the robust schedule's certified worst case is 2 to 7% below that of the nominal schedule at no measurable cost in realized AoII, and SENTINEL combined with a state-aware transmission trigger comes within 10 to 20% of the best per-slot heuristic while retaining the certificate. Without failure feedback every policy degrades by an order of magnitude and the hybrid is 25 to 30% better than the per-slot heuristic at the larger network sizes. The quantum step is exact through 25 UEs in simulation with four to five times fewer qubits than a direct epigraph encoding, and thirteen domains of 8 to 38 qubits sampled on two IBM Heron processors all decoded to the exact optimum, with the fraction of usable shots falling from 86% to 2.6% over that range. At these sizes the quantum sampler is correct but not a better optimizer than greedy association, and we report this limit together with the scale at which it would matter.**

***Index Terms*—6G, age of incorrect information (AoII), alternating direction method of multipliers (ADMM), mission-critical edge computing, non-terrestrial networks, quantum approximate optimization algorithm (QAOA), remote estimation, robust optimization, satellite-ground integrated network (SAGIN)**

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I. Introduction

Sixth-generation networks are expected to place decision-making at the edge [1]: an edge server that monitors a hazard indicator, the integrity of a structure, or the position of an asset must act on its own estimate of that process, often without a human in the loop. What matters to such a server is whether its estimate is still correct. Terrestrial mobile edge computing provides capacity and low delay where ground infrastructure exists, which is often not where monitoring is most needed, and satellite links extend coverage to the remaining regions at the price of intermittent connectivity [2]. The combination, a satellite-ground integrated network (SAGIN), is the setting of this paper.

Satellite links fail in ways that terrestrial links rarely do. A low-earth-orbit satellite sets and hands the user over to the next one every few minutes; buildings and foliage block the line of sight for the length of a pass; shadow fading in the non-line-of-sight state exceeds ten decibels [3]. During any of these interruptions the monitored process keeps evolving, and by the time the link is restored the edge estimate may be wrong by several states rather than simply old. The age of information (AoI) [4], [5] measures only elapsed time and cannot tell these two situations apart. The age of incorrect information (AoII) [6], [7] can: it is zero while the estimate is correct and grows with the duration and the magnitude of a mismatch. AoII has been analyzed mainly for one source and one monitor on a point-to-point channel. Carrying it to a network of edge nodes, satellites, and base stations raises two difficulties. The metric depends on the whole trajectory of the source since the estimate was last correct, so it does not fit the per-slot cost structure of the assignment problems used for SAGIN resource allocation. And the duration of a mismatch is governed by outages whose length is uncertain, so the allocation must be judged by its worst case over plausible disruptions, not by its average.

We treat the problem as robust combinatorial optimization. The first step is a renewal-interval decomposition: a source that evolves as a Markov chain with known transition matrix accumulates, between two deliveries, an expected AoII that depends only on the length of the interval, and this quantity can be tabulated offline. The channel then enters the

objective through a single number per scenario, the inter-delivery interval, and the resulting robust scheduling problem, VANGUARD, is a minimax assignment problem over binary variables. The second step concerns how such a problem can be given to a quantum solver. A direct encoding of the minimax, with an epigraph variable and one slack register per scenario, needs three to four auxiliary qubits for every decision qubit and a penalty landscape in which classical and quantum local search alike fail to find a feasible point; we document this failure in Section VI. SENTINEL instead dualizes the minimax into scenario weights that a classical outer loop updates by exponentiated gradient, so that each inner subproblem has a linear objective and is encoded as a compact QUBO over the decision variables and a few capacity-slack bits. The one-hot constraint of each user is enforced by the XY mixer of the quantum circuit rather than by a penalty [8], which keeps every sample a valid assignment. The network is partitioned into satellite-ground domains that are coordinated by the sharing form of ADMM [9], and each assembled schedule is evaluated exactly under every scenario, so the reported worst case is a certificate that does not rest on the solver's optimality.

The third step is an evaluation that a reader can trust. Satellite passes are propagated from a Starlink two-line-element snapshot with SGP4 [10]; the large-scale channel follows the suburban tables of 3GPP TR 38.811 [3], including the elevation-dependent line-of-sight probability and clutter loss; every policy, including the baselines, receives acknowledgement feedback and may abandon a link that has failed for a fixed time; an integer linear program solved to optimality provides the reference for every solver; and all policies run on the same random episodes, so that comparisons are paired and are reported with signed-rank tests. The quantum step is run on two IBM Heron processors as well as in simulation.

The results support a modest and precisely stated set of claims. Under real passes the worst-case certificate held in 98.7% of orchestration epochs, and its violations are non-line-of-sight passes whose clutter loss exceeds the scenario set. The robust schedule's certificate is 2 to 7% lower than that of its nominal counterpart in 67 to 100% of seeds, while the two are statistically indistinguishable in realized AoII, with and without failure feedback; the value of the scenario set is the guarantee, not a lower average. A per-slot policy that transmits whenever its observed AoII is positive is the strongest heuristic when feedback is available, and we show that it coincides with the Whittle-index scheduler of the restless-bandit literature [11], [12] at the bandwidths of this study; SENTINEL combined with the same trigger comes within 10 to 20% of it while keeping the certificate, and is 25 to 30% better than it at 15 to 20 users when feedback is removed. The compact formulation is exact through 25 users and within 12% at 50 with four to five times fewer qubits than the epigraph encoding; its limit at 75 to 100 users is the coordination of shared satellites, not the inner solver. On hardware, thirteen domains of 8 to 38 qubits sampled on `ibm_kingston` and `ibm_fez` all decoded to the exact optimum, with the fraction of usable shots decreasing from 86% to 2.6% as the circuits grew; a trained proximal-policy-optimization agent, included because learning-based baselines are expected, matches the nominal schedule at five users and falls behind as its action space grows. On every instance that fits a current device, greedy association is already optimal, so the quantum step is a correct, hardware-runnable constrained sampler and not yet a better optimizer; the instances on which greedy loses need one to two hundred qubits per domain.

The contributions of this paper are the following.

- The robust scheduling problem VANGUARD, which minimizes the worst-case time-average AoII of a network of monitored edge nodes over a scenario set of satellite shadowing and handover disruptions, jointly over update rates, association, and bandwidth, subject to capacity, visibility, rate, and AoII-budget constraints.
- A renewal-interval decomposition that expresses the expected AoII accumulated between deliveries in closed form as a function of the interval length and the source's transition matrix, verified against Monte Carlo simulation to within one percent, which reconciles the history dependence of AoII with binary-variable optimization.
- SENTINEL, a distributed hybrid procedure in which the minimax is dualized into classically updated scenario weights, each domain's subproblem is a compact QUBO with XY-mixer-enforced one-hot structure, candidate pruning removes dominated combinations without changing the optimum, and the sharing form of ADMM with a warm start and a cost-scaled penalty coordinates the domains; together with a certificate of the worst-case AoII of every schedule.
- An evaluation under Starlink ephemerides and the TR 38.811 channel, against an exact integer-programming reference, with paired statistics over 30 to 60 seeds, baselines that include the Whittle-index scheduler and a trained learning agent, an ablation of failure feedback, sensitivity to the scenario set and to bandwidth, and runs of the quantum step on two superconducting processors, with the scale limit of the quantum step stated explicitly.

The remainder of the paper is organized as follows. Section II reviews related work. Section III presents the system model: the channel and satellite geometry, the monitored sources, the disruption scenarios, the renewal-interval decomposition, the binary decision model, and the orchestration architecture. Section IV formulates VANGUARD and its binary reformulations. Section V develops the dualized compact QUBO, the XY-mixer QAOA, and the ADMM coordination. Section VI reports the evaluation, and Section VII concludes.

II. State of the Art

The paper touches five bodies of work: satellite-ground integrated edge computing, information freshness and correctness, robust resource allocation, learning-based scheduling, and quantum-assisted combinatorial optimization. We take them in turn and then state the gap.

1) Satellite-Ground Integrated Edge Computing: Mobile edge computing [13] has been extended beyond terrestrial coverage by placing edge servers on satellites and base stations [14]–[16]. The literature is dominated by task offloading under deterministic or average channel assumptions [17], [18],

with the completion delay of a task as the objective. The correctness of what the server believes about the monitored process has not been the quantity optimized. On the channel side, the 3GPP study on non-terrestrial networks [3] provides elevation-dependent line-of-sight probabilities, clutter losses, and shadow-fading statistics for several environments, and the SGP4 propagator with public two-line elements [10] gives pass geometry for any constellation; most SAGIN evaluations use neither, and we found that the choice matters, because with real passes the dominant disruptions are handovers and non-line-of-sight passes rather than coverage gaps.

2) Age of Information and Age of Incorrect Information: AoI has been studied extensively as a timeliness measure [4], [5], [12], [19], [20]. It is content-agnostic. AoII [6] penalizes the duration and magnitude of estimation error and has been analyzed for a single source under threshold and semi-Markov policies [7], [21] and over unreliable, power-constrained channels [22]. For scheduling several sources over shared channels the standard tool is the Whittle index of the restless bandit [11], applied to AoI by Kadota et al. [12]; it remains the reference policy against which new AoII schedulers should be measured, and we include it. What is missing is a formulation in which association, bandwidth, and update rates are chosen jointly across heterogeneous serving nodes, under outages that span several source transitions, with a guarantee on the worst case.

3) Robust Resource Allocation Under Link Uncertainty: Robust optimization minimizes cost against the worst realization of an uncertain parameter in a bounded set [23], [24]. When the uncertainty set is a finite list of scenarios, the minimax can be written as a maximization over scenario weights in the simplex, and the saddle point can be approached by multiplicative-weights or exponentiated-gradient updates on the weights [25]; this is the mechanism we use to keep the quantum subproblem linear. Robust allocation in satellite-ground edge computing has treated the uncertainty as a perturbation of rate or delay, not as the driver of estimation error.

4) Learning-Based Scheduling: Deep reinforcement learning [26], [27] has been applied to AoI- and AoII-aware scheduling in point-to-point and small multi-user systems. It optimizes an expected return, needs training on representative disruption patterns, and provides no bound on the estimation error under disruptions outside its training distribution. These properties do not disqualify it as a baseline, and we train one with the same reward, feedback, and episodes as every other policy; they do explain why a certified worst case is a different kind of guarantee.

5) Quantum-Assisted Combinatorial Optimization: QUBO formulations [28], [29] solved by the quantum approximate optimization algorithm (QAOA) [30], [31] or by annealing [32] have been proposed for channel assignment and scheduling in wireless networks [33]. Two developments bear directly on constrained assignment problems. The alternating-operator form of QAOA [8] replaces the transverse-field mixer by one that preserves a constraint subspace, so that one-hot and cardinality constraints need no penalty terms; and the concentration of optimal QAOA parameters across instances [34] allows angles optimized on a small instance to be reused on a larger one that cannot be simulated. On present devices, error suppression by dynamical decoupling [35] and randomized compiling [36] is inexpensive, and we report its effect on our circuits. QAOA has not previously been applied to a freshness or correctness objective, and reports of its use for network resource allocation rarely state at what size the quantum sampler stops being competitive with a classical heuristic on the same instances.

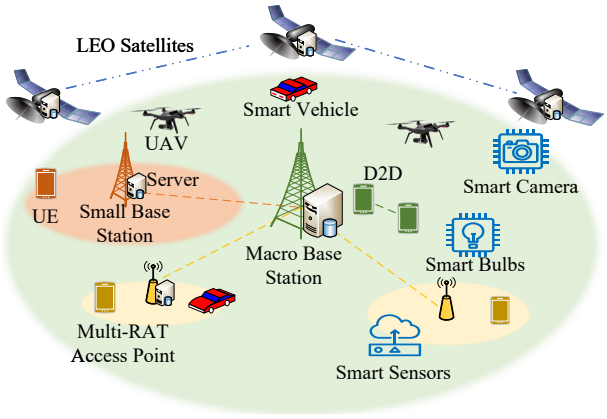


Fig. 1. Correctness-aware monitoring over a satellite-ground integrated network under satellite disruption.

6) Research Gap: No existing work formulates AoII as a robust minimax multi-node scheduling problem with joint association and resource allocation, targets information correctness rather than delay in robust SAGIN allocation, or applies quantum-assisted optimization to a correctness objective with an encoding that a current device can execute. SENTINEL addresses these three points, and its evaluation is designed so that the reader can see where each of its components helps and where it does not.

III. System Model

We consider a satellite-ground integrated edge network in which monitoring edge nodes (UEs) observe time-varying physical processes and transmit status updates to edge servers hosted at terrestrial base stations (BSs) or at visible low-earth-orbit satellites (SATs). Let $\mathcal{U}$, $\mathcal{S}$, and $\mathcal{B}$ denote the sets of UEs, SATs, and BSs, indexed by u_i, s_j, and b_n, and let $\mathcal{M} = \mathcal{S} \cup \mathcal{B}$ be the set of serving nodes. Each serving node hosts an edge server that keeps a running estimate of every process assigned to it and acts on that estimate. Time is slotted with slot duration T_s. UEs have low mobility over a decision interval; the satellite topology changes continuously because of orbital motion. Table 1 lists the notation.

In contrast to the task-offloading models that dominate SAGIN edge computing, the quantity of interest here is the correctness of the server's estimate. It depends on the dynamics of the monitored process, on when the UE samples and transmits, and on whether the transmission is delivered before the process moves again. Satellite handover and shadowing make the last of these uncertain, and an allocation chosen under nominal channel conditions can leave the server with an estimate that is wrong rather than late.

TABLE 1. Key Notations

Symbol	Definition
$\mathcal{U}$, $\mathcal{S}$, $\mathcal{B}$	Sets of UEs, SATs, and BSs
$\mathcal{M} = \mathcal{S} \cup \mathcal{B}$	Set of serving nodes
$\mathcal{K}$, $\mathcal{D}_k$	Set of domains; k-th domain
α_{ij}, ω_{in}	Association of u_i with s_j, b_n
β_{ij}^{us}, β_{in}^{ub}	Bandwidth allocated to u_i by s_j, b_n
B_j^s, B_n^b	Bandwidth capacity of s_j, b_n
$R_{ij}^{us}(\ell)$, R_{in}^{ub}	Uplink rate of u_i to s_j (scenario ℓ), to b_n
$V_{ij}(t)$, Θ	Elevation of s_j at u_i; minimum elevation
G^u, G^s, G^b	Antenna gains of the UE, SAT, and BS
λ^{cl}, σ_{sf}	TR 38.811 clutter loss; shadow-fading deviation
$\mathscr{L}$, Ω	Scenario set (L scenarios); uncertainty set
$T^{\mathrm{dis}}(\ell)$, $\tau_{imq}(\ell)$	Disruption window; induced outage (slots)
T_s, D_i	Slot duration; status-update size (bits)
$X_i(t)$, $\hat{X}_i(t)$	Monitored source of u_i; edge estimate
$\mathcal{X}_i$, P_i, π_i	State space, transition matrix, stationary distribution
$f_i(x, \hat{x})$	Distortion function
$\Delta_i^{II}(t)$, $\Delta_i^{II,\max}$	Instantaneous AoII; AoII budget
$t_{i,\kappa}$, $Y_{i,\kappa}$	κ-th delivery slot; inter-delivery interval
$\Psi_i(Y)$	Expected AoII accumulated over an interval of length Y
ρ_i, $\bar{\rho}_i^h$	Update rate of u_i; h-th rate level (H_i levels)
$\bar{\beta}_m^q$	q-th bandwidth level of node m (Q_m levels)
x_{imqh}	1 if u_i uses node m, bandwidth level q, rate level h
$\mathscr{A}_i$, $\mathscr{C}_i$	Admissible combinations of u_i; pruned candidates
$\psi_{imqh}^{(\ell)}$	Time-average AoII cost under scenario ℓ
$\mathbf{w}$, η	Scenario weights; worst-case AoII (epigraph)
T^{f}, T^{b}	Failure-detection window; blocking duration (slots)
z_{jk}, u_{jk}, ϱ	ADMM bandwidth target, scaled dual, penalty

The network is partitioned into K local satellite-ground domains $\mathcal{D}_k = (\mathcal{U}_k, \mathcal{S}_k, \mathcal{B}_k)$, $k \in \mathcal{K} = \{1, \ldots, K\}$, where the UE subsets partition $\mathcal{U}$ and a SAT may be visible to several domains. Each domain has a classical controller that builds and solves a local robust binary problem with a hybrid QAOA solver; a classical coordinator reconciles the bandwidth of SATs shared across domains and certifies every assembled network schedule by its exact worst-case AoII.

A. Satellite Geometry and Communications Model

Satellite positions are obtained from a constellation ephemeris. In the evaluation we propagate a public two-line element set with SGP4 [10], but the model only requires, for every SAT s_j and UE u_i, the slant range $d_{ij}^{us}(t)$ and the elevation angle $V_{ij}(t)$ in each slot. A SAT can serve u_i in slot t only if $V_{ij}(t) \geq \Theta$. Each UE is served by the SAT that it has been handed to and keeps that SAT until it sets below Θ, at which point the UE is handed over to the SAT with the highest elevation; a dense constellation therefore produces a sequence of passes of several minutes each, separated by handover interruptions, and no coverage gaps at the elevation masks considered here.

The nominal path loss (in dB) between u_i and s_j is

$$\varphi_{ij}^{us} = \varsigma^{us} + \zeta^{us} \log_{10}\left(f_c^{us}\right) + \xi^{us} \log_{10}\left(d_{ij}^{us}\right) + \lambda^{\mathrm{atm}}(V_{ij}), \quad (1)$$

where f_c^{us} is the carrier frequency, ς^{us}, ζ^{us}, and ξ^{us} are the reference loss and the frequency and distance exponents of free-space propagation, and $\lambda^{\mathrm{atm}}(V)$ is the atmospheric absorption, which scales with $1/\sin V$. The UE-BS path loss φ_{in}^{ub} uses the ground distance d_{in}^{ub} and a terrestrial exponent $\xi^{ub} > 2$ with a fixed margin. The received power at the SAT is $p_i + G^u + G^s - \varphi_{ij}^{us}$ in dBm, where p_i is the UE transmit power and G^u and G^s are the antenna gains of the UE terminal and the satellite; the BS link uses G^b in place of G^s and the UE's sub-6-GHz gain. The nominal uplink rate from u_i to s_j with bandwidth β_{ij}^{us} is

$$R_{ij}^{us} = \beta_{ij}^{us} \min\left\{\log_2\left(1 + \frac{10^{(p_i + G^u + G^s - \varphi_{ij}^{us})/10}}{N_0\, \beta_{ij}^{us}\, F^s}\right),\ \bar{\gamma}\right\}, \quad (2)$$

where N_0 is the noise power spectral density, F^s the receiver noise figure, and $\bar{\gamma}$ the spectral efficiency of the highest modulation and coding scheme. The UE-BS rate R_{in}^{ub} is defined in the same way. Each serving node divides its bandwidth among the UEs it serves, so the rate of a link does not depend on the association decisions of other UEs once the bandwidth level is fixed. We treat co-channel interference as absorbed in the noise figure and do not model residual Doppler, which the NTN frequency pre-compensation removes to within the subcarrier spacing.

Large-scale channel. On top of the nominal loss, each satellite link carries a large-scale state drawn once per pass from the TR 38.811 model [3]: with the elevation-dependent probability $p_{\mathrm{LOS}}(V)$ of the chosen environment the pass is in line of sight; otherwise it is in non-line-of-sight and carries the clutter loss $\lambda^{\mathrm{cl}}(V)$ of the carrier band, which at Ku band and suburban elevations is 17 to 25 dB. A shadow-fading term with the state-dependent standard deviation $\sigma_{\mathrm{sf}}(V)$ is drawn per pass, and a small-scale fading term per slot. In the suburban environment used in Section VI roughly one pass segment in twenty is in non-line-of-sight, and such a pass blocks the link for its whole duration at any bandwidth level.

B. Monitored Source and Estimation Model

Each UE u_i observes a process $X_i(t)$ modeled as a discrete-time Markov chain on a finite state space $\mathcal{X}_i$ with transition matrix P_i, in which the states are ordered severity or status levels. The UE controls a generation indicator $g_i(t) \in \{0, 1\}$ that specifies whether an update is sampled and transmitted in slot t. We assume that T_s exceeds the transmission and propagation delay of a successful update, so that an update sampled at the start of slot t is delivered within that slot or lost. Let $t_{i,\kappa}$ be the κ-th slot in which an update from u_i is delivered and $Y_{i,\kappa} = t_{i,\kappa+1} - t_{i,\kappa}$ the inter-delivery interval. The server assigned to u_i holds the estimate

$$\hat{X}_i(t) = X_i(t_{i,\kappa}), \qquad t_{i,\kappa} \leq t < t_{i,\kappa+1}, \quad (3)$$

which is fixed between deliveries while the process keeps moving. With $\Lambda_i(t) = \max\{\tau \leq t : X_i(\tau) = \hat{X}_i(\tau)\}$ the most recent slot in which the estimate was correct, the instantaneous AoII is [6]

$$\Delta_i^{II}(t) = \left(t - \Lambda_i(t)\right) f_i\left(X_i(t), \hat{X}_i(t)\right), \quad (4)$$

where $f_i(x,\hat{x}) \geq 0$ with $f_i(x,x) = 0$ measures the severity of a mismatch; we use $f_i(x,\hat{x}) = |x - \hat{x}|$. AoII returns to zero without a delivery whenever the process revisits the estimated state, and becomes positive right after a delivery if the process leaves the delivered state.

C. Disruption Scenarios

The orchestrator plans against a finite scenario set $\mathscr{L} = \{1,\dots,L\}$, each scenario ℓ specifying an additional loss $\delta^{us}_{ij}(\ell)$ on every UE-SAT link and a window length $T^{\mathrm{dis}}(\ell)$, in slots, over which it applies:

$$\tilde{\varphi}^{us}_{ij}(\ell) = \varphi^{us}_{ij} + \delta^{us}_{ij}(\ell). \tag{5}$$

The scenario rate $R^{us}_{ij}(\ell)$ follows from (2) with $\tilde{\varphi}^{us}_{ij}(\ell)$ in place of φ^{us}_{ij}, and an update sampled in slot t is delivered under scenario ℓ only if $R^{us}_{ij}(\ell)\,T_s \geq D_i$ and $V_{ij}(t) \geq \Theta$, where D_i is the size of an update in bits. BS links are terrestrial and unaffected.

We use five scenarios. The first is the nominal channel. The second is a deep shadowing event in which every satellite link loses σ_{sf} of the non-line-of-sight state, 12 dB in the suburban tables, for a window of T^{dis} slots; it stands for the shadow fading of a pass that enters non-line-of-sight. The third and fourth are the handover interruption of each individual satellite, a complete outage of that link for the duration of a handover; and the fifth is the joint outage of all satellites for the deep-shadowing window, the global worst case. An earlier version of the set also contained a mild 6 dB shadowing scenario. The sensitivity study of Section VI shows that it never binds: with the antenna gains of Table I the satellite links keep a margin above the delivery threshold at 6 dB, so no schedule changes when that scenario is added or removed, and we omit it. The study also shows that the certified worst case and the hedging behaviour of the robust schedule appear only once the handover of the second shared satellite is in the set, which is why the handover scenarios are kept per satellite rather than merged.

In an offloading model a scenario perturbs the rate. Here it also determines the length of the outage window and therefore the inter-delivery interval $Y_{i,\kappa}(\ell)$, the number of slots from $t_{i,\kappa}$ until the next slot in which $g_i(t) = 1$ and the delivery condition holds. A longer window lets $X_i(t)$ pass through more states before the next delivery. The scenario set is the planner's model of the disruption process; the simulator of Section VI draws its disruptions from the TR 38.811 state process and the ephemeris, and the agreement between the two is what the certificate's validity measures.

D. Renewal-Interval AoII Decomposition

Because $\Delta^{II}_i(t)$ depends on the trajectory of $X_i(t)$ since the estimate was last correct, it is not a per-slot function of the channel state and cannot be used directly as the cost coefficient of a binary variable. The delivery process is, however, a renewal process. Under the same-slot delivery assumption the estimate is exact at each delivery, so $\Delta^{II}_i(t_{i,\kappa}) = 0$, and between two deliveries the estimate is fixed at $\hat{x} = X_i(t_{i,\kappa})$ while the AoII evolves under the source's own dynamics, independently of the channel.

Let $\tilde{P}_{i,\hat{x}} = P_i\big(\mathbf{I} - \mathbf{e}_{\hat{x}}\mathbf{e}_{\hat{x}}^{\top}\big)$ be the taboo matrix with the $\hat{x}$-th column of P_i set to zero, so that $[\tilde{P}^{a}_{i,\hat{x}}]_{\hat{x}y}$ is the probability of moving from $\hat{x}$ to y in a steps without returning to $\hat{x}$, and let $\mathbf{f}_{i,\hat{x}}$ be the vector with entries $f_i(y,\hat{x})$. The AoII s slots after a delivery equals a $f_i(y,\hat{x})$ exactly when the source was last at $\hat{x}$ at slot $s-a$ and has since moved to y without returning, so

$$\mathbb{E}\big[\Delta^{II}_i(t_{i,\kappa}+s)\,\big|\,\hat{x}\big] = \sum_{a=1}^{s} a\,\big[P_i^{s-a}\big]_{\hat{x}\hat{x}}\big[\tilde{P}^{a}_{i,\hat{x}}\,\mathbf{f}_{i,\hat{x}}\big]_{\hat{x}}, \tag{6}$$

and the expected AoII accumulated over an interval of length Y is

$$\begin{aligned}\Psi_i(Y \mid \hat{x}) &= \sum_{s=1}^{Y-1}\mathbb{E}\big[\Delta^{II}_i(t_{i,\kappa}+s)\,\big|\,\hat{x}\big],\\ \Psi_i(Y) &= \sum_{\hat{x}\in\mathscr{X}_i}\pi_i(\hat{x})\,\Psi_i(Y \mid \hat{x}),\end{aligned} \tag{7}$$

where π_i is the stationary distribution of P_i. Averaging over π_i is legitimate because the update decisions do not depend on the realized source state, so the delivered state is stationary. By the renewal-reward theorem the time-average AoII of u_i when its deliveries are Y slots apart is $\Psi_i(Y)/Y$. Both quantities depend only on Y and P_i and are tabulated once per UE for $Y \leq Y^{\max}$; we verified the table against Monte Carlo simulation of the source and delivery process over 1.5×10^6 slots, with agreement within one percent at every interval length used.

Equation (7) separates the source from the channel. The only channel-dependent quantity the planner must reason about is the inter-delivery interval, which under scenario ℓ is deterministic for a given allocation and is denoted $Y_i(\ell)$; the cost is the tabulated ratio $\Psi_i(Y_i(\ell))/Y_i(\ell)$.

E. Binary Decision Model

The update rates and bandwidths are continuous. We discretize the bandwidth of each node $m \in \mathscr{M}$ into Q_m levels $\{\bar{\beta}^1_m,\dots,\bar{\beta}^{Q_m}_m\}$ and the update rate of u_i into H_i levels $\{\bar{\rho}^1_i,\dots,\bar{\rho}^{H_i}_i\} \subset [\rho^{\min}_i, \rho^{\max}_i]$, and define

$$x_{imqh} = \begin{cases} 1, & \text{if } u_i \text{ uses node } m, \text{ level } q, \text{ rate } h,\\ 0, & \text{otherwise.}\end{cases} \tag{8}$$

The original decisions are recovered as $\alpha_{ij} = \sum_{q,h} x_{ijqh}$, $\omega_{in} = \sum_{q,h} x_{inqh}$, $\beta^{us}_{ij} = \sum_{q,h}\bar{\beta}^q_j\,x_{ijqh}$, $\beta^{ub}_{in} = \sum_{q,h}\bar{\beta}^q_n\,x_{inqh}$, and $\rho_i = \sum_{m,q,h}\bar{\rho}^h_i\,x_{imqh}$. The scenario does not index the variable, because it is realized by the environment rather than chosen; each variable instead carries L precomputed cost coefficients. After the inadmissible combinations are removed (Section IV-A) the number of decision variables is

$$N_x = \sum_{i\in\mathscr{U}}|\mathscr{A}_i| \;\leq\; \sum_{i\in\mathscr{U}}\sum_{m\in\mathscr{M}} Q_m H_i, \tag{9}$$

which grows linearly with the number of UEs, nodes, and levels. Section V prunes each $\mathscr{A}_i$ further to a candidate set $\mathscr{C}_i$ of the κ combinations with the smallest worst-case cost, which in our instances never changed the optimum and reduced the qubit count by a further factor of two to three.

F. Delivery Feedback and Link-Failure Handling

A serving node acknowledges each delivered update, so a UE knows at the end of a slot whether its transmission succeeded. A UE whose transmissions to a node have failed in every slot of a window of T^{f} slots declares the link failed and avoids that node for T^{b} slots or, for a satellite, until its next handover; its controller is informed and re-solves the domain problem without the failed link at the next orchestration interval. This rule is applied uniformly to every policy in Section VI, because any state-aware baseline needs the same information to be competitive, and because the value of a robust schedule depends on whether such feedback exists. Both regimes are reported.

G. Distributed Orchestration Model

Each domain $\mathcal{D}_k$ holds the local decision vector $\mathbf{x}_k = \{x_{imqh} \mid u_i \in \mathcal{U}_k, (m,q,h) \in \mathcal{C}_i\}$ and the local state $\mathbf{s}_k = [\mathbf{\Psi}_k, \mathbf{V}_k, \mathbf{X}_k, \mathbf{B}_k, \mathbf{E}_k, \mathbf{\Xi}_k]$, with $\mathbf{\Psi}_k$ the tabulated AoII costs of its UEs, $\mathbf{V}_k$ the elevations, $\mathbf{X}_k$ the transition matrices and budgets, $\mathbf{B}_k$ the bandwidth levels, $\mathbf{E}_k$ the ephemeris over the decision interval, and $\mathbf{\Xi}_k$ the scenario set restricted to the domain. The controller re-solves its problem whenever a satellite becomes visible or sets, when a link failure is reported, and in any case at a fixed re-solve period. Within one solve, a classical outer loop maintains scenario weights, each domain solves a linear assignment subproblem on a quantum sampler, and the coordinator runs the sharing form of ADMM on the bandwidth of shared satellites and evaluates the worst-case AoII of each assembled schedule exactly. The orchestration latency, the time from the start of a solve until its schedule is in force, extends every inter-delivery interval by ϵ_i slots and enters the cost through $\Psi_i(Y_i(\ell) + \epsilon_i)$; in the evaluation the solve time is measured and reported rather than assumed.

IV. Problem Formulation

The objective is to minimize the worst-case time-average AoII across all monitored UEs over the scenario set, by jointly choosing update rates, association, and bandwidth. Let $\boldsymbol{\chi}_i = \big(\rho_i, \{\alpha_{ij}\}_j, \{\omega_{in}\}_n, \{\beta^{us}_{ij}\}_j, \{\beta^{ub}_{in}\}_n\big)$ collect the decisions of u_i and $Y_i(\boldsymbol{\chi}_i, \boldsymbol{\xi})$ its inter-delivery interval under realization $\boldsymbol{\xi} \in \Omega$. The robust Value-of-information-aware, uncertainty-robust, Network-wide Generation, Update-scheduling, Association, and Resource orchestration under Disruption (VANGUARD) problem is

$$
\begin{aligned}
\mathscr{P}_1 : \min_{\{\boldsymbol{\chi}_i\}_{i\in\mathcal{U}}} \max_{\boldsymbol{\xi}\in\Omega} \; & \frac{1}{|\mathcal{U}|} \sum_{i\in\mathcal{U}} \frac{\Psi_i\big(Y_i(\boldsymbol{\chi}_i,\boldsymbol{\xi})\big)}{Y_i(\boldsymbol{\chi}_i,\boldsymbol{\xi})} \\
\text{s.t. } \; & \mathrm{C1} : \sum_{j\in\mathcal{S}} \alpha_{ij} + \sum_{n\in\mathcal{B}} \omega_{in} = 1, \; \forall\, i \in \mathcal{U}, \\
& \mathrm{C2} : \sum_{i\in\mathcal{U}} \alpha_{ij}\, \beta^{us}_{ij} \le B^s_j, \; \forall\, j \in \mathcal{S}, \\
& \mathrm{C3} : \sum_{i\in\mathcal{U}} \omega_{in}\, \beta^{ub}_{in} \le B^b_n, \; \forall\, n \in \mathcal{B}, \\
& \mathrm{C4} : \frac{\Psi_i\big(Y_i(\boldsymbol{\chi}_i,\boldsymbol{\xi})\big)}{Y_i(\boldsymbol{\chi}_i,\boldsymbol{\xi})} \le \Delta_i^{II,\max}, \; \forall\, i \in \mathcal{U}, \; \forall\, \boldsymbol{\xi} \in \Omega, \\
& \mathrm{C5} : \alpha_{ij}\big(V_{ij} - \Theta\big) \ge 0, \; \forall\, i \in \mathcal{U}, \; \forall\, j \in \mathcal{S}, \\
& \mathrm{C6} : \rho_i^{\min} \le \rho_i \le \rho_i^{\max}, \; \forall\, i \in \mathcal{U}, \\
& \mathrm{C7} : \alpha_{ij}, \, \omega_{in} \in \{0,1\}, \; \forall\, i, j, n.
\end{aligned}
$$

C1 routes each UE through exactly one serving node per decision interval. C2 and C3 are the bandwidth capacities. C4 is the correctness constraint: the expected time-average AoII of every UE stays within its budget $\Delta_i^{II,\max}$ under every realization, including the worst. It is stated in expectation because the instantaneous AoII of a Markov source cannot be bounded deterministically; the source may stay in a mismatched state for a whole interval. C5 restricts association to visible satellites, C6 bounds the update rate by the freshness requirement and the UE's energy or channel-access budget, and C7 makes association binary.

$\mathscr{P}_1$ is a mixed-integer nonlinear minimax program. With a single scenario and fixed rates it contains the generalized assignment problem and is NP-hard. The inner maximization adds a difficulty of its own: because Ψ_i is a nonlinear, source-specific function of the interval, the realization that maximizes the network AoII is not necessarily the one with the worst channel, since it also depends on how the volatility of each source interacts with the outage that the realization induces. The feasible association set defined by C5 changes at every decision interval as satellites move.

A. Robust Binary Reformulation

With the discretization of Section III-E and the variable x_{imqh} of (8), each scenario ℓ gives, for every UE-node pair and bandwidth level, a rate $R_{imq}(\ell)$ from (2) with the scenario path loss. The link is in outage throughout the scenario window if that rate cannot deliver an update within a slot, so the induced outage is

$$
\tau_{imq}(\ell) = \begin{cases} T^{\mathrm{dis}}(\ell)\, \mathbb{I}\big\{R_{imq}(\ell)\, T_s < D_i\big\}, & m \in \mathcal{S}, \\ 0, & m \in \mathcal{B}. \end{cases} \tag{10}
$$

The bandwidth level matters: a larger allocation can keep the rate above the delivery threshold through deep shadowing, whereas a handover interruption is an outage at any bandwidth. A UE whose update fails during an outage transmits a fresh sample as soon as the link returns. The worst case is an outage that begins at a scheduled sampling slot, which gives

$$
Y_{imqh}(\ell) = \left\lceil \frac{1}{\bar{\rho}_i^h T_s} \right\rceil + \tau_{imq}(\ell), \qquad \psi^{(\ell)}_{imqh} = \frac{\Psi_i\big(Y_{imqh}(\ell)\big)}{Y_{imqh}(\ell)}, \tag{11}
$$

precomputed for every (i,m,q,h,ℓ); with orchestration overhead, $Y_{imqh}(\ell) + \epsilon_i$ replaces $Y_{imqh}(\ell)$.

Pre-elimination. The admissible set $\mathscr{A}_i$ consists of the combinations (m,q,h) for which (i) $m \in \mathcal{B}$, or $m \in \mathcal{S}$ with $V_{im} \ge \Theta$; (ii) the nominal rate satisfies $R_{imq} T_s \ge D_i$; and (iii) $\max_\ell \psi^{(\ell)}_{imqh} \le \Delta_i^{II,\max}$. Condition (i) enforces C5 and (iii) is the discrete form of C4. Since each UE selects exactly one combination, it meets its budget under every scenario if and only if its selection lies in $\mathscr{A}_i$; the budget is thus enforced by removing variables rather than by adding penalty terms, which costs no qubits. A UE for which $\mathscr{A}_i$ is empty, which happens for a satellite-only UE whose budget is below the AoII of a

full handover outage, is assigned its least-bad combination and reported as a budget violation; Section VI states how often this occurs.

The binary reformulation of VANGUARD is

$$\begin{aligned}
\mathscr{P}_2 :\ & \min_{\mathbf{x}} \ \max_{\ell\in\mathscr{L}} \ \frac{1}{|\mathscr{U}|}\sum_{i\in\mathscr{U}}\sum_{(m,q,h)\in\mathscr{A}_i} \psi^{(\ell)}_{imqh}\, x_{imqh} \\
\text{s.t. } & \text{D1}: \sum_{(m,q,h)\in\mathscr{A}_i} x_{imqh} = 1,\ \forall\, i\in\mathscr{U}, \\
& \text{D2}: \sum_{i\in\mathscr{U}}\sum_{(j,q,h)\in\mathscr{A}_i} \bar{\beta}^q_j\, x_{ijqh} \le B^s_j,\ \forall\, j\in\mathscr{S}, \\
& \text{D3}: \sum_{i\in\mathscr{U}}\sum_{(n,q,h)\in\mathscr{A}_i} \bar{\beta}^q_n\, x_{inqh} \le B^b_n,\ \forall\, n\in\mathscr{B}, \\
& \text{D4}: x_{imqh}\in\{0,1\},\ \forall\, i\in\mathscr{U},\ (m,q,h)\in\mathscr{A}_i.
\end{aligned}$$

D1 is the one-hot selection, D2 and D3 the discrete capacities, D4 the binary domain.

B. Epigraph Form, Certificate, and Exact Reference

Introducing the worst-case value η gives the equivalent epigraph form

$$\begin{aligned}
\mathscr{P}_3 :\ & \min_{\mathbf{x},\,\eta}\ \eta \\
\text{s.t. } & \text{E1}: \frac{1}{|\mathscr{U}|}\sum_{i\in\mathscr{U}}\sum_{(m,q,h)\in\mathscr{A}_i} \psi^{(\ell)}_{imqh}\, x_{imqh} \le \eta,\ \forall\, \ell\in\mathscr{L}, \\
& \text{D1–D4},
\end{aligned}$$

in which E1 is tight for at least one scenario at the optimum. $\mathscr{P}_3$ serves two purposes in this paper, neither of which is to be solved on a quantum device. First, it is the certificate: for any schedule $\mathbf{x}$ returned by any solver, the L left-hand sides of E1 are evaluated exactly, so the worst-case time-average AoII of that schedule over $\mathscr{L}$ is known with certainty. Second, $\mathscr{P}_3$ is a mixed-integer linear program, and up to 75 UEs it is solved to proven optimality by a branch-and-cut solver [37] in well under a second, with a time limit reached only on some 100-UE instances; this gives an exact reference for every heuristic and quantum solver we compare. A direct QUBO encoding of $\mathscr{P}_3$, with η on a binary grid and a slack register for each of the L constraints E1, was the encoding of an earlier version of this work; it multiplies the qubit count by three to four and produces a penalty landscape in which neither simulated annealing nor QAOA found a feasible assignment on instances of ten UEs. Section V therefore works with $\mathscr{P}_2$ directly, replacing the inner maximization by scenario weights so that every quantum subproblem is linear in $\mathbf{x}$ and the only auxiliary qubits are the slack bits of the capacity constraints D2 and D3.

V. Proposed Solution

Problem $\mathscr{P}_2$ is a robust assignment problem whose only coupling across UEs is the capacity of the serving nodes. SENTINEL solves it in three steps. The minimax over scenarios is handled by a classical outer loop on scenario weights, so that every subproblem handed to a quantum sampler has a linear objective. Each domain's subproblem is encoded as a compact QUBO over its decision variables and a few capacity-slack bits, with the one-hot constraint enforced by the mixer of the quantum circuit rather than by a penalty. The domains are coordinated through the sharing form of ADMM. Every schedule produced at any point of the procedure is certified exactly by its worst-case AoII over $\mathscr{L}$, so the procedure can be stopped at any time with a valid guarantee in hand.

A. Scenario-Weight Dualization of the Minimax

Write $c^{(\ell)}(\mathbf{x}) = \frac{1}{|\mathscr{U}|}\sum_i \sum_{(m,q,h)\in\mathscr{A}_i} \psi^{(\ell)}_{imqh} x_{imqh}$ for the network AoII under scenario ℓ. For any $\mathbf{x}$ and any weight vector $\mathbf{w}$ in the simplex $\Delta_L = \{\mathbf{w}\ge 0,\ \sum_\ell w_\ell = 1\}$,

$$\max_{\ell\in\mathscr{L}} c^{(\ell)}(\mathbf{x}) = \max_{\mathbf{w}\in\Delta_L} \sum_{\ell\in\mathscr{L}} w_\ell\, c^{(\ell)}(\mathbf{x}), \tag{12}$$

so $\mathscr{P}_2$ is $\min_{\mathbf{x}} \max_{\mathbf{w}\in\Delta_L} \mathbf{w}^\top \mathbf{c}(\mathbf{x})$. We approach the saddle point by alternating an inner step on $\mathbf{x}$ for fixed $\mathbf{w}$, which is the step handed to the quantum sampler, with an exponentiated-gradient step on $\mathbf{w}$ [25]:

$$\mathbf{x}^{(t)} \in \arg\min_{\mathbf{x}\ \text{s.t. D1–D4}} \sum_\ell w^{(t)}_\ell\, c^{(\ell)}(\mathbf{x}), \tag{13}$$

$$\eta^{(t)} = \max_\ell c^{(\ell)}(\mathbf{x}^{(t)}), \tag{14}$$

$$w^{(t+1)}_\ell \propto w^{(t)}_\ell \exp\!\big(\sigma\, c^{(\ell)}(\mathbf{x}^{(t)})/\eta^{(t)}\big), \tag{15}$$

with step $\sigma > 0$ and $\mathbf{w}^{(1)}$ uniform, for T outer iterations. The weights move toward the scenarios that hurt the current schedule most, and the next inner solve is pulled toward schedules that do better on them. Equation (14) is the exact worst-case AoII of $\mathbf{x}^{(t)}$ over $\mathscr{L}$, computed in $O(L|\mathscr{U}|)$; SENTINEL returns the iterate with the smallest $\eta^{(t)}$ together with that value as its certificate. Because the inner problem (13) has a linear objective, it needs neither the epigraph variable nor the scenario slacks of a direct minimax encoding. In our instances those auxiliary registers, $M_\eta + \sum_\ell W_\ell$ qubits per domain, outnumbered the decision qubits.

B. Candidate Pruning

The pre-elimination of Section IV-A removes inadmissible combinations. We further retain, for each UE, only the κ admissible combinations with the smallest worst-case cost $\max_\ell \psi^{(\ell)}_{imqh}$, plus the best combination at every other visible node so that the capacity coupling is preserved, and denote the result $\mathscr{C}_i \subseteq \mathscr{A}_i$. The number of decision qubits per UE is then at most $\kappa + |\mathscr{M}|$ instead of $\sum_m Q_m H_i$. Pruning could in principle discard an optimal combination that is individually poor but relieves a congested node; with $\kappa = 4$ this did not happen on any of the instances of Section VI, where the exact optimum over $\mathscr{C}$ equalled the exact optimum over $\mathscr{A}$ from 10 to 100 UEs.

C. Compact Domain QUBO

For domain k with weights $\mathbf{w}$ and ADMM state $(\mathbf{z}_k, \mathbf{u}_k)$, let $a_v = \frac{1}{|\mathscr{U}|}\sum_\ell w_\ell \psi^{(\ell)}_v$ for $v = (i,m,q,h)$, $i\in\mathscr{U}_k$, $(m,q,h)\in\mathscr{C}_i$, and let $b_m(\mathbf{x}_k) = \sum_{i\in\mathscr{U}_k}\sum_{(m,q,h)\in\mathscr{C}_i} \bar{\beta}^q_m x_{imqh}$ be the bandwidth that domain k uses at node m. Satellites

visible only to domain k form $\mathcal{S}_k^{\mathrm{o}}$; those shared with other domains form $\mathcal{S}_k^{\mathrm{sh}}$. The binary vector $\mathbf{y}_k = (\mathbf{x}_k, \boldsymbol{\nu}_k)$ holds the decision variables and, for each node $m \in \mathcal{S}_k^{\mathrm{o}} \cup \mathcal{B}_k$, a capacity slack $\nu_m = \Delta_\beta \sum_b 2^b y_{m,b}$ of $W_m = \lceil \log_2(B_m/\Delta_\beta + 1) \rceil$ bits, where the bandwidth levels and capacities are integer multiples of Δ_β. The domain QUBO is

$$F_k(\mathbf{y}_k) = \sum_v a_v x_v + \lambda \sum_{m \in \mathcal{S}_k^{\mathrm{o}} \cup \mathcal{B}_k} \big(b_m(\mathbf{x}_k) + \nu_m - B_m\big)^2 + \frac{\varrho}{2} \sum_{j \in \mathcal{S}_k^{\mathrm{sh}}} \big(b_j(\mathbf{x}_k) - z_{jk} + u_{jk}\big)^2. \quad (16)$$

The first term is the weighted AoII cost, the second enforces the capacity of the nodes that belong to the domain alone, and the third is the augmented-Lagrangian coordination term for the shared satellites, with penalty parameter ϱ. Constraint D1 does not appear: the mixer of Section V-D keeps every UE group one-hot, so on the subspace the circuit can reach, $x_v x_{v'} = 0$ for two candidates of the same UE and the corresponding entries of the QUBO matrix are dropped. The matrix $\mathbf{Q}^{(k)}$ collects the linear coefficients of F_k on its diagonal, since $y^2 = y$ for binary y, and the pairwise coefficients split between the two symmetric off-diagonal entries.

Proposition 1. *Let $\bar{a}_k = \sum_{i \in \mathcal{U}_k} \max_{v \in \mathcal{C}_i} a_v$ and let $\Gamma_k^{\max}$ and $\Gamma_k^{\min}$ be the largest and smallest values of the coordination term over one-hot assignments of domain k. If*

$$\lambda > \frac{\bar{a}_k + \Gamma_k^{\max} - \Gamma_k^{\min}}{\Delta_\beta^2} \quad (17)$$

and the domain problem is feasible, then every one-hot minimizer of F_k satisfies the capacity constraints of $\mathcal{S}_k^{\mathrm{o}} \cup \mathcal{B}_k$ and minimizes the weighted cost plus coordination term over the feasible one-hot assignments.

Proof. Over one-hot assignments the cost term lies in $[0, \bar{a}_k]$. For a feasible assignment whose slack bits represent the unused capacity exactly, the penalty vanishes and $F_k \le \bar{a}_k + \Gamma_k^{\max}$. For an assignment that exceeds the capacity of some node, the residual of that node is a positive integer multiple of Δ_β for every choice of slack bits, since the slack is non-negative, so the penalty is at least $\lambda \Delta_\beta^2$ and $F_k \ge \lambda \Delta_\beta^2 + \Gamma_k^{\min} > \bar{a}_k + \Gamma_k^{\max}$. No infeasible one-hot assignment is therefore a minimizer, and on the feasible set F_k equals the cost plus coordination term once the slack bits are chosen to cancel the penalty. □

The bound is taken over one-hot assignments because that is the subspace the circuit explores. Taking it over the whole hypercube, as a penalty analysis of a transverse-field QAOA would require, inflates λ by more than an order of magnitude and flattens the energy landscape to the point where neither simulated annealing nor QAOA finds a feasible point; this was the behaviour of the epigraph encoding and is quantified in Section VI-D. The number of qubits of domain k is

$$\hat{N}_k = \sum_{i \in \mathcal{U}_k} |\mathcal{C}_i| + \sum_{m \in \mathcal{S}_k^{\mathrm{o}} \cup \mathcal{B}_k} W_m, \quad (18)$$

independent of L. With the parameters of Section VI and two domains, this is 22, 34, 46 and 58 qubits per domain at 5, 10, 15 and 20 UEs, against 120 to 260 for the epigraph encoding of the same instances (Fig. 2).

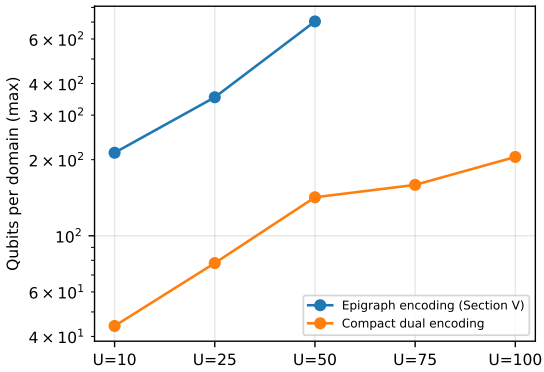


Fig. 2. Qubits per domain of the compact formulation (18) and of the direct epigraph encoding of $\mathscr{P}_3$ (one epigraph register and one slack register per scenario) for the instances of Section VI.

D. Ising Mapping and XY-Mixer QAOA

The QUBO is mapped to an Ising Hamiltonian through the substitution $y_a = (1 - \sigma_a^z)/2$, where a indexes the $\hat{N}_k$ qubits of the domain and σ_a^z is the Pauli-Z operator on qubit a. Discarding additive constants,

$$H_C^{(k)} = \sum_a h_a \sigma_a^z + \sum_{a<a'} J_{aa'} \sigma_a^z \sigma_{a'}^z, \qquad h_a = -\frac{1}{2} \sum_{a'} Q_{aa'}^{(k)}, \qquad J_{aa'} = \frac{1}{2} Q_{aa'}^{(k)}, \quad (19)$$

where the sums over a' run over the symmetric matrix $\mathbf{Q}^{(k)}$. Couplings $J_{aa'}$ arise only between candidates that use the same node, and between those candidates and the node's slack bits, so the cost layer is sparse: the candidates of one UE are not coupled to each other, and candidates at different nodes are not coupled at all.

For the mixer we do not use the transverse field $\sum_a \sigma_a^x$, which would leave the one-hot subspace and force D1 back into the cost as a penalty. Instead, on each UE group we apply a ring of XY interactions and keep the transverse field only on the slack bits [8]:

$$H_M^{(k)} = \sum_{i \in \mathcal{U}_k} \sum_{(a,a') \in \mathrm{ring}(\mathcal{C}_i)} \tfrac{1}{2}\big(\sigma_a^x \sigma_{a'}^x + \sigma_a^y \sigma_{a'}^y\big) + \sum_{a \in \mathrm{slack}} \sigma_a^x, \quad (20)$$

where ring$(\cdot)$ denotes adjacent pairs in a cyclic ordering of the group. Each XY term conserves the Hamming weight of its group, so every Trotter factor of $e^{-\mathrm{i}\theta H_M^{(k)}}$ does as well, and a state initialized in the one-hot subspace stays in it exactly; the only way a measured sample can violate D1 is through hardware noise, which the decoder detects. Each group is initialized either in its W state, the uniform superposition of its one-hot basis states, prepared with a linear-depth ladder of controlled rotations, or in the basis state of the greedy

assignment of Section V-E, which we call the warm start. The depth-p state is

$$|\boldsymbol{\gamma},\boldsymbol{\theta}\rangle = \prod_{l=1}^{p} e^{-\mathrm{i}\theta_l H_M^{(k)}} e^{-\mathrm{i}\gamma_l H_C^{(k)}} |\psi_0\rangle, \tag{21}$$

and the angles $(\boldsymbol{\gamma},\boldsymbol{\theta})$ are optimized by a derivative-free classical optimizer on the sampled energy $\langle H_C^{(k)}\rangle$. The sampled bitstrings are decoded classically; samples that are not one-hot or exceed a capacity are discarded, and the feasible sample with the smallest value of (16) is returned as $\hat{\mathbf{x}}_k$. When $\hat{N}_k$ exceeds the capacity of the available device or simulator, the inner step is performed by a one-hot-preserving simulated annealer on the same QUBO, which moves one UE's selection at a time; Section VI reports which solver produced each result.

E. Distributed Coordination

The domains are coupled through the satellites they share. For a shared satellite s_j, let $\mathscr{K}_j \subseteq \mathscr{K}$ be the K_j domains that see it. The coordinator assigns each $k \in \mathscr{K}_j$ a bandwidth target z_{jk} and keeps a scaled dual variable u_{jk}; the local QUBOs carry the coordination term of (16). After the domains return $\hat{\mathbf{x}}_k^{(r+1)}$ in round r, the coordinator sets $\tilde{a}_{jk} = b_j(\hat{\mathbf{x}}_k^{(r+1)}) + u_{jk}^{(r)}$ and applies the sharing-form ADMM updates [9]

$$z_{jk}^{(r+1)} = \tilde{a}_{jk} - \frac{1}{K_j}\max\Big\{0,\ \sum_{k'\in\mathscr{K}_j}\tilde{a}_{jk'} - B_j^s\Big\}, \tag{22}$$

$$u_{jk}^{(r+1)} = u_{jk}^{(r)} + b_j\big(\hat{\mathbf{x}}_k^{(r+1)}\big) - z_{jk}^{(r+1)}, \tag{23}$$

which lower every target at an overloaded satellite by an equal share of the excess and accumulate the remaining violation in the duals. The round ends when the primal residual $\big(\sum_{j,k}(b_j(\hat{\mathbf{x}}_k^{(r+1)}) - z_{jk}^{(r+1)})^2\big)^{1/2}$ and the dual residual $\varrho\big(\sum_{j,k}(z_{jk}^{(r+1)} - z_{jk}^{(r)})^2\big)^{1/2}$ fall below a tolerance or after $R_{\max}$ rounds; ϱ is adapted by residual balancing.

Two details decide whether this loop produces good schedules, and both were learned from failures. First, the targets are initialized from an uncoupled round ($\varrho = 0$), projected onto capacity, and not from the equal share B_j/K_j. With equal-share targets and the two-sided quadratic coordination term, every domain is pulled toward using exactly its share of every shared satellite whether or not it needs it, and the loop declares convergence after two rounds at a schedule that we measured to be 60 to 560% above the optimum at 50 to 100 UEs. Second, ϱ is scaled to the mean per-UE spread of the weighted costs a_v, so that one bandwidth unit of deviation from the target costs about one typical per-UE cost difference; a penalty two orders of magnitude larger than the cost differences, which is what a generic choice produced, makes the coordination term the only thing the inner solver sees. If no capacity-feasible network schedule has been found when the loop stops, a greedy repair moves UEs from overloaded satellites to the candidate with the smallest increase in worst-case cost at a node with spare capacity. The complete procedure is given in Algorithm 1.

Convergence and certification. For convex local problems the sharing form of ADMM converges [9]; here the local problems are binary, so the loop is a heuristic and neither it nor the outer weight iteration is guaranteed to reach the saddle point of (12). SENTINEL does not rely on either for correctness. The incumbent is replaced only by network schedules that satisfy D1 to D3, and its worst-case AoII (14) is computed exactly, so the number of iterations affects the quality of the returned schedule but never the validity of its certificate.

Algorithm 1: SENTINEL: Distributed Robust Orchestration for VANGUARD

Input : $\mathscr{U}, \mathscr{S}, \mathscr{B}$, $\{P_i\}$, capacities, budgets, levels, Θ, $\mathscr{L}$, κ, T, σ, p, $R_{\max}$, ϵ_{tol}.
Output: $\mathbf{x}^\star$ decoded into ρ_i, α_{ij}, ω_{in}, β_{ij}^{us}, β_{in}^{ub}; certificate $\eta^\star$; worst scenario $\ell^\star$.

Partition the network into K domains; identify shared SATs.
Tabulate $\Psi_i(\cdot)$ by (7), the costs $\psi_{imqh}^{(\ell)}$ by (11), the admissible sets $\mathscr{A}_i$ and the pruned candidates $\mathscr{C}_i$.
$\mathbf{w}^{(1)} \leftarrow$ uniform; $\eta^\star \leftarrow +\infty$.
for $t = 1, \dots, T$ **do**
  Warm start: solve each domain with $\varrho = 0$; set $z_{jk}^{(0)}$ to the projected loads, $u_{jk}^{(0)} \leftarrow 0$; scale ϱ to the cost spread.
  for $r = 0, \dots, R_{\max} - 1$ **do**
    for *each domain* k in parallel **do**
      Build F_k by (16) with $\mathbf{w}^{(t)}$, $\mathbf{z}_k^{(r)}$, $\mathbf{u}_k^{(r)}$; map to $H_C^{(k)}$ by (19).
      if $\hat{N}_k$ *fits the sampler* **then** run XY-mixer QAOA (21), decode, keep the best feasible sample as $\hat{\mathbf{x}}_k^{(r+1)}$;
      else obtain $\hat{\mathbf{x}}_k^{(r+1)}$ by one-hot-preserving annealing of F_k;
    Update $z_{jk}^{(r+1)}$, $u_{jk}^{(r+1)}$ by (22)–(23); adapt ϱ.
    if $\hat{\mathbf{x}}^{(r+1)}$ *satisfies D1–D3 and* $\max_\ell c^{(\ell)}(\hat{\mathbf{x}}^{(r+1)}) < \eta^\star$ **then**
      $\mathbf{x}^\star \leftarrow \hat{\mathbf{x}}^{(r+1)}$; $\eta^\star \leftarrow \max_\ell c^{(\ell)}(\mathbf{x}^\star)$.
    if *residuals* $\le \epsilon_{\mathrm{tol}}$ **then break**;
  if *no feasible schedule in this iteration* **then** apply greedy repair and evaluate (14);
  $\mathbf{x}^{(t)} \leftarrow$ best schedule of this iteration; update $\mathbf{w}^{(t+1)}$ by (15).
Decode $\mathbf{x}^\star$; $\ell^\star \leftarrow \arg\max_\ell c^{(\ell)}(\mathbf{x}^\star)$.

F. Resource Scaling and Complexity

Qubits and gates. By (18) the qubit count of a domain grows with $\kappa|\mathscr{U}_k|$ and the slack registers, and does not depend on L. Because the cost Hamiltonian couples only candidates that share a node, the cost layer of domain k needs at most $\sum_m \binom{n_m+W_m}{2}$ two-qubit ZZ rotations, where n_m is the number of candidates at node m in the domain and $W_m = 0$ for shared satellites; the XY mixer adds $|\mathscr{C}_i|$ two-qubit rotations per UE group and layer, and the W-state preparation $|\mathscr{C}_i| - 1$. On

a heavy-hex processor, routing a domain of 18 to 24 qubits yielded 58 to 305 two-qubit gates after transpilation at depth 1, and 606 gates for a 38-qubit domain (Section VI-E). The direct epigraph encoding, whose scenario penalties couple every pair of local variables, needs $\hat{N}_k(\hat{N}_k-1)/2$ rotations per layer with $\hat{N}_k$ three to five times larger.

Time. The offline tabulation of Ψ_i costs $O(Y^{\max}|\mathcal{X}_i|^3 + |\mathcal{X}_i|(Y^{\max})^2)$ per UE and the coefficients $\psi^{(\ell)}_{imqh}$ cost $O(L\sum_i|\mathcal{A}_i|)$ lookups, once per set of sources and scenarios. Assembling and mapping F_k costs $O(\hat{N}_k^2)$. If the angle optimizer runs I_{QAOA} iterations with N_{shot} shots each, the quantum work of domain k per round is $O(I_{\mathrm{QAOA}}\, p\, N_{\mathrm{shot}}\, G_k)$ with G_k the gate count above, the ADMM updates cost $O(\sum_j K_j)$ per round, and the certificate costs $O(L|\mathcal{U}|)$. With the domains run in parallel, the wall-clock cost of a solve is

$$C_{\mathrm{SENTINEL}} = O\Big(T\, R_{\max}\big(I_{\mathrm{QAOA}}\, p\, N_{\mathrm{shot}} \max_k G_k + L|\mathcal{U}|\big)\Big). \quad (24)$$

In the evaluation, a solve of a 20-UE network with $T = 6$ and $R_{\max} = 20$ took 1 to 7 seconds on the classical annealer and the orchestration loop used 34 to 223 seconds of solver time per 10-minute run, which is reported rather than assumed in the AoII figures through the overhead ϵ_i of Section III-G. Exhaustive search over $\mathscr{P}_2$ costs $O(L|\mathcal{U}|\prod_i|\mathcal{A}_i|)$, and branch-and-cut on $\mathscr{P}_3$ has the same worst case although it is fast on our instances; a centralized QAOA would need $\sum_k \hat{N}_k$ qubits on one device. The advantage of the decomposition is therefore the qubit count per device, not asymptotic time, and Section VI reports at which size the inner sampler and the coordination each stop being exact.

VI. Performance Evaluation

We evaluate SENTINEL with the simulator released with this paper[1] under the parameters of Table 2, which are the values used for every figure. Each configuration runs for 60,000 slots of 10 ms, ten minutes of network time, on 30 independent seeds; the policies of one seed share its topology, source trajectories, fading, and disruption realization, so all comparisons are paired. We report medians with interquartile ranges, because a few seeds contain long non-line-of-sight passes that dominate means; means with 95% confidence intervals, the raw per-seed results, and the paired test tables are in the repository.

Baselines. *Random* transmits with probability 0.1 per slot over a uniformly chosen visible node with equal bandwidth sharing. *AoII threshold* is a state-aware policy that transmits whenever its AoII is positive, at most once per 5 slots, on the node with the best nominal rate. *Whittle index* is the restless-bandit scheduler of the AoII literature [11], [12]: the index of each UE's (source state, estimate, age) arm is computed numerically per source by relative value iteration on a grid of transmission prices, and every slot the mismatched UEs are ranked by index and admitted at their best node while the node's shared bandwidth still delivers every admitted packet; it has the threshold policy's observations and rate cap. *PPO* is an actor-critic agent [27] that selects, once per orchestration interval, a serving node and an update-period level for every UE, with invisible or failed nodes masked and equal bandwidth sharing, rewarded with the negative mean AoII of the interval; it is trained for 5000 episodes per network size on 30 seeds disjoint from the evaluation seeds and evaluated deterministically on the same episodes as every other

[1] https://github.com/tanzimulalam/Quantum_SAGIN_AoII_Updated_Simulation, tag `v2.1-manuscript`; the script `scripts/reproduce_all.sh` regenerates every table and figure.

TABLE 2. Simulation Parameters (values used for all figures)

Parameter	Value
Network	
Number of UEs	$\{5, 10, 15, 20\}$; up to 100 in Sec. VI-D
BSs / satellite serving slots	2 / 2 (topology study: 0 to 3 each)
Area; BS height; UE height	20×20 km^2; 25 m; 1.5 m
Satellite passes	Starlink two-line-element snapshot propagated with SGP4 over the MTSU site (35.85°N, 86.39°W); minimum elevation 25°
Domains K	2 (one BS each; satellites shared); K = number of BSs in Sec. VI-D
Channel and resources	
UE transmit power	23 dBm
UE–BS: carrier, bandwidth, path loss	3.5 GHz, 20 MHz, $-147.55 + 20\log_{10} f + 26\log_{10} d$ + 6 dB, BS gain 15 dBi, noise figure 7 dB
UE–SAT: carrier, bandwidth, path loss	12 GHz, 50 MHz, free space +1 dB, gains 30 dBi (UE) + 30 dBi (SAT), noise figure 9 dB
NTN large-scale model	TR 38.811 suburban: LOS probability, clutter loss, shadow fading per pass; atmospheric 0.5 dB/sin V
Bandwidth levels $\bar{\beta}^q_m$ ($Q_m = 3$)	$\{2.5, 5, 10\}$ MHz, $\Delta_\beta = 2.5$ MHz
Spectral-efficiency cap	6 (BS), 4 (SAT) bit/s/Hz
Source and AoII	
Source states; transition matrix	4; birth–death, self-transition $\mathcal{U}[0.90, 0.99]$ per UE
Distortion; AoII budget $\Delta_i^{II,\max}$	$\lvert x - \hat{x}\rvert$; 10
Update periods ($H_i = 4$); slot; packet	$\{5, 10, 20, 50\}$ slots; 10 ms; 20 kbit
Disruption	
Scenario set ($L = 5$)	nominal; deep shadowing 12 dB for 30 slots; handover of each SAT (20 slots); joint loss (30 slots)
Realized process (per 1 s interval)	TR 38.811 LOS/NLOS state and shadow fading per pass; shadowing event per UE–SAT link w.p. 0.07; handover failure per SAT w.p. 0.03; natural handovers at pass changes
Link-failure handling (all policies)	node blocked after 60 consecutive failed slots, for 6 s or until the satellite's next handover; SENTINEL re-solves without the link
Solver	
Formulation	compact dual, $\kappa = 4$, $T = 6$ outer iterations, $\sigma = 2$
Local solver	XY-mixer QAOA, depth 1, 512 shots, COBYLA 40 iterations (domains $\leq$ 26 qubits); one-hot-preserving annealing otherwise
ADMM ϱ_0, tolerance, rounds	scaled to cost spread, 10^{-2}, 20
Re-solve	on every satellite visibility change, on reported link failure, and at least every 30 s
Quantum backend	Qiskit Aer (noiseless) [38]; FakeTorino / FakeFez noise models; `ibm_kingston`, `ibm_fez` (Heron r2)

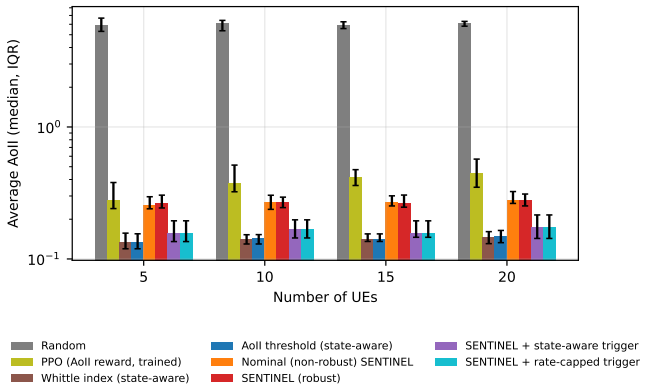


Fig. 3. Median average AoII (bars) and interquartile range (whiskers) over 30 seeds versus the number of UEs, with Starlink passes, the TR 38.811 channel, and the stochastic disruption process active.

TABLE 3. Median Average AoII with Starlink Passes and the TR 38.811 Channel (30 seeds, disruption on)

UEs	Random	PPO	Whittle	AoII thr.	Nominal	SENTINEL	+ trigger
5	5.94	0.280	0.134	0.134	0.253	0.265	0.155
10	6.06	0.372	0.141	0.142	0.270	0.269	0.168
15	5.87	0.413	0.142	0.142	0.269	0.264	0.155
20	6.05	0.446	0.146	0.149	0.276	0.278	0.175

policy. *Nominal SENTINEL* runs Algorithm 1 with the nominal scenario alone ($L = 1$). *SENTINEL* is the robust schedule with the five-scenario set. *SENTINEL + trigger* uses SENTINEL's association and bandwidth with transmissions triggered by the observed mismatch, rate-capped at the chosen period. Every policy receives the same acknowledgement feedback and applies the same link-failure rule of Section III-F. The exact reference for Section VI-D is the integer linear program $\mathscr{P}_3$ solved with HiGHS [37].

A. AoII Versus Network Size

Fig. 3 and Table 3 give the median average AoII from 5 to 20 UEs. Three groups of policies are visible. The state-aware per-slot policies, the AoII threshold and the Whittle index, are the lowest at 0.13 to 0.15 and coincide with each other; Section VI-C explains why. The SENTINEL schedules, robust and nominal, sit at 0.25 to 0.28 and do not grow with the number of UEs, because the bandwidth levels are dedicated and a re-solve redistributes load as satellites come and go. SENTINEL with the state-aware trigger lies between the two groups at 0.155 to 0.175: it keeps the robust association and bandwidth, transmits only when the estimate is wrong, and thereby raises the informative-update ratio from 0.23 to 0.96 while retaining the certificate of the schedule it runs on. Random scheduling is at 6, and the trained PPO agent matches the nominal schedule at 5 UEs (0.280 against 0.253) and degrades to 0.446 at 20 UEs as its action space grows; it routes all traffic to satellites and, like SENTINEL, samples without observing the source, so it never reaches the state-aware group.

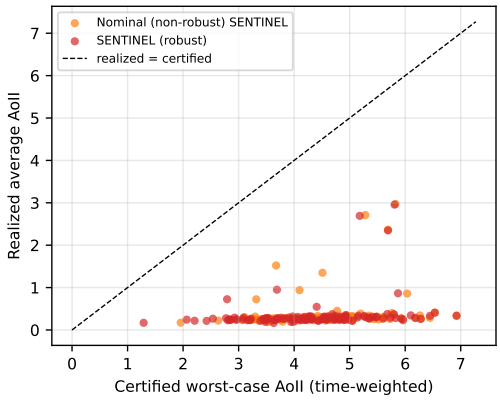


Fig. 4. Certified worst-case AoII of each orchestration run against its realized average AoII, robust and nominal SENTINEL, all network sizes and seeds of Fig. 3. Points below the diagonal are runs in which the realized average stayed within the certificate.

The other metrics follow the same pattern. Delivery ratios are 0.92 to 0.94 for every policy except random, and the per-slot policies deliver with 2.8 to 3.0 ms of end-to-end latency against 4.7 to 5.8 ms for the SENTINEL schedules, whose dedicated bandwidth levels are smaller than an equal share of an otherwise idle node. The satellite tier carries 73 to 79% of robust SENTINEL's deliveries, 77 to 86% of the nominal schedule's, 97 to 98% of the threshold policy's, and all of PPO's. Peak AoII is 100 to 400 for every policy and is set by the 20-slot handover interruptions and the non-line-of-sight passes rather than by the scheduler. A solve of the 20-UE network used 20 to 58 qubits per domain and the orchestration loop consumed 34 to 223 s of solver time per ten-minute run, counted in the AoII through ϵ_i.

B. The Certificate and the Value of Robustness

Fig. 4 plots the certified worst case of every run against the AoII it realized. The realized average stayed within the certificate in 98.7% of orchestration epochs for the robust schedule and 98.5% for the nominal one. The violations are passes that enter the non-line-of-sight state, whose 17 to 25 dB clutter loss exceeds the 12 dB deep-shadowing scenario; with a synthetic pass model that produces coverage gaps longer than any scenario window the figure was 98.0%, and every violation there was such a gap. The certificate is therefore valid against the scenario set it was computed from, and the paper states its scope that way.

What the scenario set buys is visible in the certificate itself and not in the realized AoII. Across the 30 paired seeds, the robust schedule's certified worst case is 2 to 7% below the nominal schedule's (paired-median ratio 0.93 at 5 UEs to 0.98 at 20 UEs), lower in 67 to 100% of seeds, with Wilcoxon signed-rank $p \le 9 \times 10^{-5}$ at every size after Holm correction. The realized AoII of the two schedules is statistically indistinguishable at every size ($p \ge 0.08$), although the robust schedule's mean at 20 UEs is lower, 0.39 against 0.46, because it keeps 21 to 27% of deliveries on base

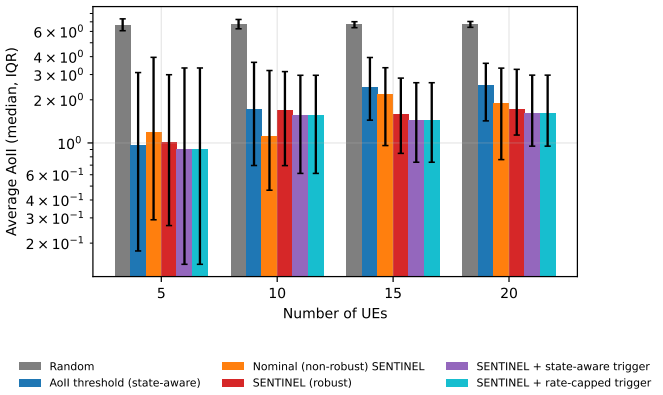


Fig. 5. Median average AoII (60 seeds) when acknowledgement feedback and link-failure handling are disabled for every policy. All policies degrade by an order of magnitude; the per-slot threshold policy loses its advantage.

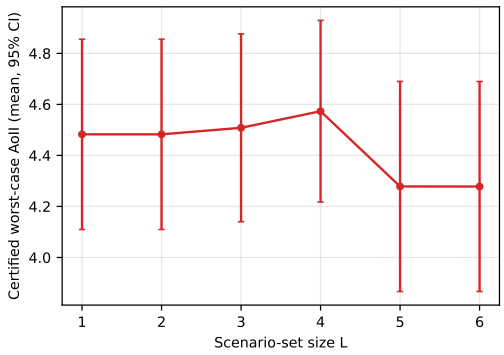


Fig. 6. Certified worst-case AoII of the robust schedule at 10 UEs as the scenario list (nominal, mild, deep, handover of satellite 1, handover of satellite 2, joint) is truncated to its first L members; mean and 95% confidence interval over 30 seeds.

stations that no satellite disruption can reach. The gap between SENTINEL with the trigger and the threshold policy, 0.02 to 0.03 in AoII, is small but significant at every size ($p \leq 0.013$), and PPO is significantly behind SENTINEL from 10 UEs on.

Without failure feedback. Whether the network can react to a dead link matters more than any scheduling choice. Fig. 5 repeats the sweep with the link-failure rule disabled for every policy, on 60 seeds because the resulting distributions are heavy-tailed. Median AoII rises by an order of magnitude for all policies, the certificate holds in 87 to 92% of epochs, and the ranking changes: the threshold policy, which concentrates 98% of its traffic on the best nominal satellite, ties the SENTINEL schedules at 5 and 10 UEs and falls 25 to 30% behind SENTINEL with the trigger at 15 and 20 UEs (Hodges-Lehmann differences −0.77 and −0.71, Holm-corrected $p = 0.023$ and 0.029, the hybrid lower in 70 to 72% of seeds), while the SENTINEL schedules keep 18 to 27% of deliveries on base stations. Robust and nominal SENTINEL remain indistinguishable in realized AoII in this regime as well ($p \geq 0.41$). We draw the conclusion plainly: the measurable product of the robust scenario set is its lower certificate, obtained in both regimes, and the measurable value of a scheduled base-station share appears when feedback is absent.

Scenario-set size. To see which scenarios matter, the list was extended by a mild 6 dB shadowing scenario and truncated to $L = 1, \ldots, 6$ at 10 UEs (Fig. 6). Realized AoII is flat in L (0.265 to 0.278). The certificate and the base-station hedge move only once the handover of the second shared satellite enters the set: from $L = 4$ to $L = 5$ the certified worst case falls from 4.57 to 4.28 and the satellite share from 0.79 to 0.74. The mild scenario never binds: removing it from the full set leaves the certified worst case of every schedule unchanged on all 30 seeds at every network size, with realized-AoII differences below 10^{-3}, which is why the paper's scenario set has five members. Nominal plus the joint outage alone would reproduce the same schedule at a third of the dual iterations.

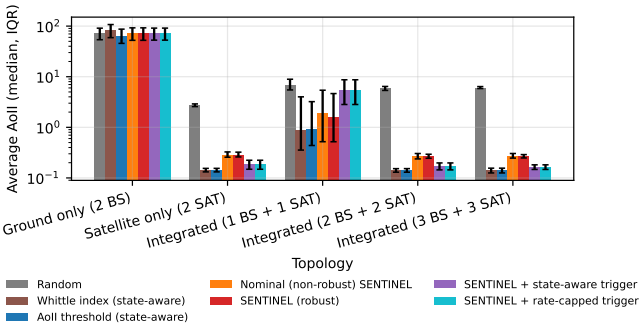


Fig. 7. Median average AoII at 10 UEs for ground-only, satellite-only, and integrated deployments (30 seeds, disruption on, logarithmic scale).

C. Topology, the Whittle Index, and Capacity

Fig. 7 compares deployments at 10 UEs. Two base stations cover only the UEs within 5 to 7 km of the 20×20 km^2 area, so ground-only monitoring fails for every policy (median AoII 63 to 81). Two satellite serving slots alone bring the per-slot policies to 0.14 and the SENTINEL schedules to 0.18 to 0.28, and the integrated $2 + 2$ and $3 + 3$ deployments are the best for every policy, with SENTINEL routing 26 to 36% of its deliveries to base stations. The one deployment in which the SENTINEL schedules are clearly worse than the per-slot policies is a single base station with a single satellite slot (medians 1.6 to 1.9 against 0.87 to 0.90, and 5.4 for the triggered hybrid): with one slot, every natural handover is a total outage for every UE on it, and a periodic or rate-capped schedule loses more during the recovery than a policy that retransmits the moment the link returns. We report this as a limitation of scheduled transmission at that topology.

Whittle index and capacity. At the bandwidths of Table 2 a node delivers 19 to 40 simultaneous equal sharers within one slot, so capacity never binds and the Whittle scheduler

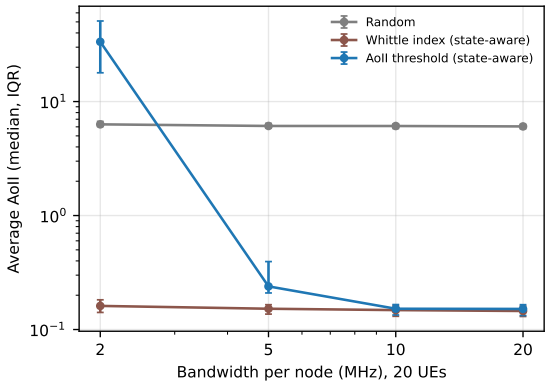


Fig. 8. Median average AoII (IQR over 30 seeds) of the per-slot state-aware policies at 20 UEs as the bandwidth per node is reduced. The Whittle-index scheduler and the AoII-threshold policy coincide while capacity does not bind and separate below 5 MHz, where uncoordinated threshold transmissions collapse the shared links.

TABLE 4. Mean Optimality Gap (%) Above the Exact ILP Optimum of $\mathscr{P}_3$

Setting	Qubits/domain	XY-QAOA	Compact ADMM	Epigraph ADMM	Greedy+LS
4–8 UEs, 2+2	8–24	0.0 (15/15 optimal)	0.0	38–51	0.0
10 / 25 UEs, 2+2	44 / 78	–	0.0	42.7 / 1.8	0.0
50 UEs, 4+3	142	–	11.5	38.8	6.0
75 / 100 UEs	159 / 205	–	49.9 / 60.5	–	14.4 / 11.5
Severity sweep, 20 UEs	29–66	–	0.0	0.2–4.7	0.0
$L = 1/3/5/6$, 20 UEs	60–77	–	0.0/3.2/6.9/0.0	74/233/9.0/3.8	0/0/1.1/0

makes the same decisions as the threshold policy; their AoII is identical to three decimals at every size and in every topology ($p \geq 0.18$), and the threshold baseline of this paper is the Whittle-index policy of this regime. Fig. 8 reduces the bandwidth per node at 20 UEs. At 5 MHz the threshold policy is 57% worse than the index scheduler (0.239 against 0.152, $p = 4 \times 10^{-9}$), and at 2 MHz its simultaneous transmissions collapse every shared link (median AoII 33.5, delivery ratio 0.16) while the capacity-aware index scheduler stays at 0.16. SENTINEL's dedicated bandwidth levels avoid this regime by construction. The index scheduler's one weakness is the ground-only deployment, where its admission rule never attempts a node that cannot deliver at the nominal rate and the threshold policy's blind attempts occasionally succeed on a fading upswing (81 against 63).

D. Validation Against the Exact Minimax Optimum

Table 4 reports the mean gap to the exact ILP optimum of $\mathscr{P}_3$ over capacity-feasible schedules (10 seeds; 5 for the QAOA rows), for the XY-mixer QAOA on the statevector simulator, for the full distributed pipeline of Algorithm 1 with the compact QUBO and with the epigraph encoding of our earlier version, and for greedy association refined by local search. Three observations follow. First, the XY-mixer QAOA returned the exact optimum on every instance whose domains fit the simulator (8 to 24 qubits), with 100% of samples one-hot and capacity-feasible; the epigraph encoding, run with the same classical search, is 38 to 51% above the optimum on the same instances and needs 133 to 193 qubits, and a plain simulated annealer on its penalty landscape did not find a single feasible assignment in 50 solves at 10 UEs. Second, the compact distributed pipeline is exact through 25 UEs and within 12% at 50 UEs; its limit at 75 to 100 UEs is the coordination of five shared satellites among six to eight domains, which neither more ADMM rounds nor smaller dual steps resolve, and where greedy association with local search remains within 11 to 14%. Third, candidate pruning with $\kappa = 4$ never changed the exact optimum. The exact reference itself was solved to proven optimality in under a second on every instance up to 75 UEs and reached its time limit on one in ten instances at 100 UEs.

TABLE 5. Transpiled Circuits ($\kappa = 4$, depth 1, greedy start) and Sampling Results on Noise Models and Devices

Instance	Dom.	Qubits	2q gates / depth	Backend	Feasible shots	Best = opt.
6 UEs, s1	0	18	137 / 169	FakeTorino (model)	52.1%	yes
6 UEs, s1	1	20	279 / 355	FakeTorino (model)	33.1%	yes
6 UEs, s1	0	18	139 / 188	FakeFez (model)	69.6%	yes
6 UEs, s1	1	20	301 / 436	FakeFez (model)	47.6%	yes
6 UEs, s1	0	18	139 / 187	ibm_kingston	63.3%	yes
6 UEs, s1	1	20	301 / 428	ibm_kingston	42.7%	yes
6 UEs, s2	0	18	139 / 187	ibm_kingston	60.9%	yes
6 UEs, s2	1	12	58 / 76	ibm_kingston	73.5%	yes
6 UEs, s3	0	24	305 / 387	ibm_kingston	34.8%	yes
6 UEs, s3	1	14	120 / 187	ibm_kingston	69.6%	yes
6 UEs, s5	0	12	58 / 74	ibm_kingston	75.9%	yes
6 UEs, s5	1	12	58 / 74	ibm_kingston	76.4%	yes
6 UEs, s6	0	8	38 / 77	ibm_fez	86.5%	yes
6 UEs, s6	1	16	79 / 77	ibm_fez	61.8%	yes
6 UEs, s7	0	12	58 / 76	ibm_fez	71.9%	yes
6 UEs, s7	1	18	142 / 171	ibm_fez	59.4%	yes
10 UEs, s1	0	38	606 / 626	ibm_kingston	2.6%	yes
6 UEs, s3	0	24	305 / 386	ibm_kingston, control	25.0%	yes
6 UEs, s3	0	24	305 / 386	ibm_kingston, DD	21.1%	yes
6 UEs, s3	0	24	305 / 386	ibm_kingston, DD + twirling	13.8%	yes
10 UEs, s1	0	38	606 / 626	ibm_kingston, control	2.4%	yes
10 UEs, s1	0	38	606 / 626	ibm_kingston, DD	0.7%	yes
10 UEs, s1	0	38	606 / 626	ibm_kingston, DD + twirling	0.2%	yes

E. Quantum Resources and Device Results

Table 5 lists the transpiled circuits of the hardware-sized domains (depth 1, greedy warm start, optimization level 3, eight routing trials) and the results of sampling them on two Heron noise models and on the ibm_kingston and ibm_fez processors (Heron r2, 156 qubits, 27 and 28 October 2026; job identifiers in the repository). Every scenario weight, ADMM target, and QAOA angle was optimized on the simulator; each device job executed only the final sampling circuit of one or two domains with 4096 shots, and the whole campaign used 47 s of processor time. The 38-qubit domain exceeds the statevector simulator, so it received the angles of the largest simulated domain of its run, relying on the concentration of QAOA parameters across instances [34]; the summary file of that run records this.

Device noise breaks the one-hot structure or the capacity of a part of the shots; the decoder discards them. In all thirteen unmitigated device domains the best surviving sample equals the noiseless optimum, and every decoded network schedule attains the exact ILP optimum of its instance. The feasible-shot fraction is the honest measure of the noise: 60 to 86% at 8 to 18 qubits with up to 139 two-qubit gates, 35 to 43% at 20 to 24 qubits with about 300 gates, and 2.6% at 38 qubits with 606 gates, bracketed by the two noise models where both are available. At the 38-qubit scale the device still contains the optimum among 4096 shots, but a few hundred more two-qubit gates would leave fewer than one feasible shot in a thousand, which places the present limit of the quantum step at about ten UEs per domain at depth 1. Error suppression does not move this limit. Re-sampling the two largest domains

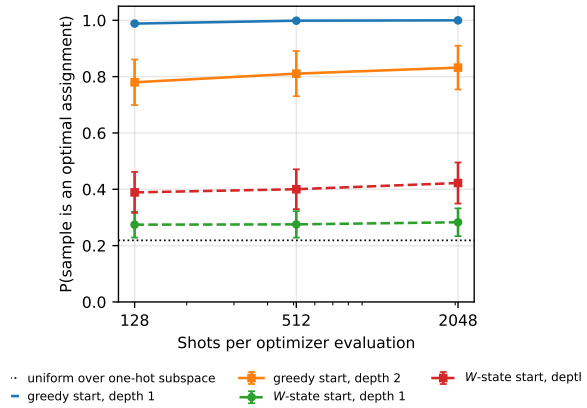


Fig. 9. Probability that a sample of the XY-mixer QAOA is an optimal assignment, on the hardware-sized domains of three 6-UE instances (12 to 24 qubits), versus circuit depth and shots per optimizer evaluation, from the W-state and the greedy warm start. The dashed level is the uniform distribution over the one-hot subspace.

in one calibration session with dynamical decoupling (XY4 sequence) [35] and with dynamical decoupling plus Pauli twirling of the two-qubit gates and measurements [36] lowered the feasible fraction from the in-session control values of 25.0% and 2.4% to 21.1% and 0.7%, and to 13.8% and 0.2%, respectively. These circuits have almost no idle time, so decoupling pulses add gate error without cancelling idle dephasing, and twirling inserts further single-qubit gates; the best sample equalled the noiseless optimum in every run regardless. The unmitigated yield of the same circuit also varied by about ten points between calibration sessions (34.8% against 25.0%), which is why every device comparison in this paper is made within one session.

What the QAOA layers contribute at this size is shown in Fig. 9. On every hardware-sized domain the greedy assignment is already optimal, and the role of the layers is to concentrate a constrained sampler: from the W state, one layer raises the probability of drawing an optimal assignment from the uniform 0.22 to 0.28 and two layers to 0.42, independently of the shot count above 128; from the greedy basis state the optimum is retained with probability 0.99 at depth 1. The optimum is contained in the samples in every run because the one-hot subspace of a hardware-sized domain has at most a few hundred states. We therefore do not claim a quantum advantage. The instances on which greedy association is suboptimal, 50 UEs and above in Table 4, need 140 to 200 qubits per domain, beyond simulation and beyond the gate budget that the feasible-shot fractions above allow on current devices; whether a sampler at that size concentrates on the optimal set is the question this work leaves open.

VII. Conclusion

This paper asked how a satellite-ground edge network should schedule status updates when what matters is whether the edge estimate is correct, and when the satellite links that carry most of the traffic fail for minutes at a time. We formulated the question as the robust problem VANGUARD, in which update rates, association, and bandwidth are chosen to minimize the worst-case time-average AoII over a set of disruption scenarios, and showed that a renewal-interval decomposition reduces the history-dependent AoII to a tabulated cost per inter-delivery interval, so that the problem becomes a minimax assignment over binary variables. SENTINEL solves it by dualizing the minimax into scenario weights that a classical loop updates, encoding each domain's linear subproblem as a compact QUBO whose one-hot structure the XY mixer of the circuit preserves, and coordinating the domains with the sharing form of ADMM; every schedule it returns carries an exactly computed worst-case certificate.

The evaluation used Starlink ephemerides, the 3GPP TR 38.811 channel, acknowledgement feedback for every policy, an exact integer-programming reference, paired tests over 30 to 60 seeds, and two IBM Heron processors. Its findings are these. The certificate held in 98.7% of orchestration epochs under real passes, and its violations are non-line-of-sight passes outside the scenario set. The robust schedule's certificate is 2 to 7% below the nominal schedule's in two thirds to all of the seeds, while the two are indistinguishable in realized AoII with and without failure feedback; the value of the scenario set is the guarantee. A per-slot policy that transmits whenever its observed AoII is positive is the strongest heuristic when feedback exists, coincides with the Whittle-index scheduler at the paper's bandwidths, and loses to the capacity-aware index scheduler when bandwidth per node is cut; SENTINEL with the same trigger comes within 10 to 20% of it while keeping the certificate, and is 25 to 30% better than it at 15 to 20 UEs when feedback is removed. The compact formulation needs four to five times fewer qubits than the direct epigraph encoding, is exact through 25 UEs, and is limited at 75 to 100 UEs by the coordination of shared satellites. Thirteen device domains of 8 to 38 qubits all decoded to the exact optimum, with the fraction of usable shots falling from 86% to 2.6% over that range, and six re-samples with dynamical decoupling and twirling lowered that fraction rather than raised it; at these sizes the quantum sampler is correct but not a better optimizer than greedy association, and the instances on which it could be require one to two hundred qubits per domain.

Three directions follow. The coordination of many domains sharing a few satellites is the bottleneck above 50 UEs, and a coordinator that negotiates per-UE assignments rather than aggregate bandwidth may remove it. The scenario set should be learned from the pass statistics of the site rather than fixed, since the handover scenario was the one that mattered and the shadowing scenario was outrun by non-line-of-sight passes. And the question of whether a constrained quantum sampler concentrates on the optimal set at the 150-qubit scale, where greedy association starts to fail, is open and will be answered on hardware rather than in simulation.

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